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# RAMON M. P. MARIANO

## IoT Systems Engineer | Full-Stack Developer

Curitiba, Paraná, Brazil | +55 41 99796-3433 | reversmon@gmail.com  
[linkedin.com/in/ramonmpm](https://linkedin.com/in/ramonmpm) | [github.com/reverstay](https://github.com/reverstay)

### PROFESSIONAL SUMMARY

I integrate software, electronics and field operations to build and support connected systems. My experience combines full-stack development, medical telemetry, technical leadership and additive manufacturing. At OPx, I contributed to telemetry growth from **10 to 72 units (+620%)** and reduced production and installation cost per module by **31.5%**. Electronic Engineering undergraduate at UTFPR, with graduation expected in 2027.

### PROFESSIONAL EXPERIENCE

#### OPx Soluções Inteligentes | Curitiba, Brazil

**Technical Supervisor** | Feb 2026 - Sep 2026

**Development Lead - Software and Electronics** | Apr 2025 - Feb 2026

**Electronic Engineering Intern - Clinical Engineering** | Feb 2025 - Apr 2025

Progressed from engineering internship to technical supervision, combining hands-on development with team coordination across electronics, software and additive manufacturing. Selected contributions across these roles:

- Contributed to telemetry expansion from **10 to 72 units (+620%)** by supporting field deployments, troubleshooting and improvements to assembly and installation processes.
- Reduced production and installation cost per module by **31.5%** by improving the production process.
- Developed **40+ spare parts for 15+ medical equipment models** from Philips, GE and Siemens through reverse engineering, 3D scanning, digital modeling and fabrication.
- Delivered a replacement link now used in **one GE abdominal MRI coil** by designing the component and coordinating manufacturing in China.
- Standardized assembly, configuration and testing guidance in **20+ process manuals** by consolidating technical procedures; supported **15+ field installations** through equipment configuration, fault diagnosis and assistance to field engineering teams.
- Contributed full-stack software and infrastructure to **Comando Remoto**, a solution developed for remote operation of **GE MRI models**.
- Built and maintained the **CLÍNICA Remota** management platform for documents, reporting, procurement, inventory, internal communication and Google-integrated authentication; developed quality reporting for equipment assembly, installation and testing.
- Established a deployment pipeline for **three internal applications** by configuring **Ubuntu Server** infrastructure and application release workflows.
- Supported monitoring equipment connectivity through **six telemetry SIMs** by arranging mobile-data services.
- Developed electronic assembly procedures and test jigs, implemented hardware and firmware improvements, and supported reverse engineering of MRI coil and preamplifier circuits.
- Produced prototypes using **PLA, ABS, TPU, nylon, PETG and PET-CF** with Blender, Autodesk Fusion 360, a Creality Raptor scanner, and Creality and Bambu Lab printers.

## ADDITIONAL PROFESSIONAL EXPERIENCE

### Copel | Electrical Engineering Intern - DMED

Jan 2023 - Oct 2024

- Automated recurring energy-metering routines and analyzed measurement data using Python, pandas, openpyxl, Excel and Power BI.
- Supported telemetering equipment replacement and communication testing at substations by accompanying field teams and checking equipment connectivity.
- Coordinated intern assignments and supported equipment assessment, supplier research and testing for energy-metering operations.

### Uaxica | Flutter Developer and Front-End Web Developer

Jan 2024 - Jun 2024 | Remote | Company based in Luanda, Angola

- Developed mobile interfaces for **two user groups - passengers and drivers** - using Flutter and Dart, including navigation, input validation, business logic and backend service integration.
- Supported localization and Android/iOS asset preparation; created Figma screen prototypes and prepared Android builds for testing with Gradle and Gradlew.
- Developed **one corporate website** for the mobility platform using Bootstrap, JavaScript and Node.js, maintaining responsive layouts and visual consistency with the mobile application.

## TECHNICAL SKILLS

**Software & APIs:** Python; Django / Django REST Framework / Channels; Flask; FastAPI; Kotlin / Quarkus; REST APIs; C / C++.

**Web & mobile:** React; TypeScript / JavaScript; Angular; HTML / CSS; Bootstrap; Tailwind CSS; shadcn/ui; Vite; Flutter / Dart; Android Studio; Xcode.

**Infrastructure:** Linux / Ubuntu Server; Docker / Docker Compose; Nginx; Git / GitHub; SSH; network and routing diagnostics; Certbot / Let's Encrypt.

**Data & automation:** PostgreSQL; SQL; relational modeling; legacy data migration; pandas; NumPy; openpyxl; Selenium; PyAutoGUI; Power BI / DAX; Excel / VBA.

**Embedded systems & electronics:** ESP32; Arduino; Raspberry Pi; ROS; FreeRTOS; PID; I2C; 2D LiDAR; PCB design with KiCad / EasyEDA; THT / SMD soldering; USB Gadget; KVM over IP.

**Design & manufacturing:** Reverse engineering; 3D scanning; additive manufacturing; Blender; Autodesk Fusion 360; Creality Raptor; Creality / Bambu Lab; G-code workflows.

**Methods & AI:** Technical leadership; technical documentation; process improvement; Kanban; Kaizen; 5S; Prompt Engineering.

## EDUCATION

### Bachelor's Degree in Electronic Engineering - In Progress

Universidade Tecnológica Federal do Paraná (UTFPR) | Expected graduation: 2027

### Technical Diploma in Electrotechnics

FORMA-BRASIL Educacional | 2024

### Technical Diploma in Logistics

Colégio Estadual Presidente Lamenha Lins | 2018

## LANGUAGES & PROFESSIONAL TRAINING

**Portuguese:** Native | **English:** Advanced | **Spanish:** Intermediate

Professional registration: Conselho Federal de Técnicos Industriais (CFT). Safety training: NR10, NR12 and NR35.